

Faculty of Engineering, Alexandria University

Communication and Electronics Department

GRADUATION PROJECT PROPOSAL

AI-Powered SOAR Platform

for Automated Cyber Defense in Banking Systems

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1. Introduction

Imagine a banking platform that does not only process transactions, but also thinks, learns, and protects itself. This project introduces an intelligent banking ecosystem that combines artificial intelligence, cybersecurity, and modern software engineering to create a safer and smarter banking experience.

From a customer's perspective, the system provides a complete digital banking environment where users can securely manage accounts, transfer money, apply for loans, and receive personalized services. Behind the scenes, AI models continuously analyze activity to detect suspicious behavior before it becomes a serious problem.

For bank employees and security teams, the platform acts as an intelligent assistant. It detects potential cyberattacks, explains security risks, and supports faster decision-making through a centralized monitoring dashboard. Instead of reacting after an incident occurs, the system helps the bank predict and prevent threats.

By bringing together banking services, artificial intelligence, and cybersecurity, the project demonstrates a vision for the future of digital banking: a system that is not only secure and efficient, but also proactive in protecting customers and supporting bank employees.

2. Project Core

2.1 System Interfaces & Access Control

The platform is built around three role-based interfaces, each governed by the principle of least privilege — every account type can access only what its role requires.

User (Customer) Interface — Least Privilege

- Make transactions (transfers, payments)
- Check account balance
- View current offers and promotions
- No administrative or cross-account data access

Employee Interface — Role-Based Access

Employees are segmented by department, and each department has its own policies and permission set, so staff only see and act on data relevant to their function:

- Human Resources (HR)
- Customer Service
- Finance & Accounting
- Information Technology (IT)
- Marketing
- Communications & PR
- Legal & Regulatory Affairs

Security Team (SOC) Interface — Full Oversight

- Full access to system and security logs
- Visibility and control over everything that affects the security posture of employees and the platform as a whole

2.2 Cybersecurity & SOC Module

Responsible for protecting the banking platform through continuous monitoring, intelligent threat detection, and automated incident response.

Security Requirements, Log Preparation & Attack Planning

- **Asset & requirement analysis:** identify critical assets and define security requirements
- **Attack modeling:** representative attacks including Brute Force, SQL Injection, Fraudulent Transactions, and Session Hijacking
- **Log collection & normalization:** ingest raw logs from system components and map them into a unified schema
- **Data contracts:** standardized JSON contracts defining how logs reach the AI detection model and how responses return to the platform

System Integration, Optimization, Testing & Evaluation

- **End-to-end integration:** Decision Engine, AI predictions, backend APIs, and SOC Dashboard operate as one workflow
- **Optimization:** refining detection thresholds, reducing false positives, updating playbooks based on test results
- **Red Team evaluation:** the Cybersecurity team simulates attacks to validate detection accuracy, playbook selection, and dashboard alerting

AI-Driven Vulnerability Assessment

- **Continuous scanning:** an AI module reviews the platform's vulnerabilities on a day-by-day basis
- **Up-to-date intelligence:** cross-references live sources to flag outdated components (e.g., an outdated PHP version) as soon as they are detected
- **Real-time alerting:** the SOC team is notified immediately of newly discovered exposures
- **Cross-interface reporting:** findings are reported through all three interfaces, scoped to what is relevant for each audience

2.3 Machine Learning / AI Module

Provides the intelligence layer of the platform: threat detection, fraud prevention, risk assessment, and behavioral analysis.

Security AI

- **Threat Detection & Classification:** core model analyzing security logs to identify malicious activity and severity
- **Anomaly Detection:** learns baseline behavior and flags significant deviations, enabling detection of unseen attacks
- **Threat Intelligence Matching:** checks IPs, domains, and file hashes against known malicious campaigns
- **Insider Threat Detection:** monitors employee behavior for excessive record access, abnormal hours, or unauthorized data exports

Banking AI

- **Loan Approval / Credit Risk Prediction:** estimates the probability an applicant fails to repay a loan
- **Transaction Fraud Detection:** analyzes individual transactions and historical behavior to estimate fraud likelihood
- **Customer Behavior Anomaly Detection:** flags new locations, new devices, and unusually large purchases
- **Multi-Account Fraud Detection (Optional):** examines relationships across accounts to surface coordinated fraud or money laundering

Shared AI Component

- **Explainability Module:** a language model that explains why the AI reached a given decision, for security analysts
- **Customer Chatbot:** a restricted, customer-facing version of the explainability model available on the User Interface to answer questions and resolve confusion in real time

2.4 Full Stack Module

Connects the user interface with business logic, the database, AI services, and the cybersecurity components.

Frontend

- **Login Page:** unified authentication that redirects each user to the correct interface based on account type
- **Customer Pages:** account management, transactions, loan applications, exchange rates, and support
- **Employee Pages:** customer management, loan processing, internal communication, and scheduling — scoped per department
- **IT/SOC Pages:** threat and anomaly feeds, incident response, audit logs, and system health monitoring
- **Shared Features:** notifications, settings, and logout, common to all authorized users

Backend

- **Authentication & Login:** validates credentials, determines account type, and issues role-based tokens
- **Customer Services:** payments/transfers (fraud-checked), loan processing (risk-checked), and exchange rate retrieval
- **Employee Services:** loan review with AI risk assessment, and secure customer database lookup
- **IT/SOC Services:** threat/anomaly feeds, incident response actions, audit logging, and access management
- **Shared Services:** real-time notifications and secure logout with token invalidation

2.5 DevOps Module

Ensures reliable data flow, efficient AI deployment, seamless service communication, and a scalable, secure, highly available platform.

Data Management & Processing Pipeline

- **Database management:** automated pipelines integrate system logs, network traffic, and banking databases with minimal delay
- **Log ingestion & normalization:** logs are parsed into a unified schema and streamed through Kafka to avoid bottlenecks

AI Deployment & System Integration

- **MLOps:** each model is containerized with Docker and orchestrated with Kubernetes for scaling and self-recovery
- **Integration layer:** Kafka topics connect AI models, the Decision Engine, and the banking backend asynchronously
- **API management:** an API Gateway handles routing, authentication, rate limiting, versioning, and traffic management

Infrastructure Operations & Security Management

- **CI/CD:** every update is packaged as a Docker image and deployed across staging and production Kubernetes environments
- **Monitoring & alerting:** tracks service availability, latency, throughput, and API error rates with automated alerts
- **Secrets management:** credentials and API keys are protected via secure secret management, network policies, and access controls

2.6 Platform Security Architecture

A defense-in-depth security design spanning the network edge, input handling, authentication, and transaction integrity.

Perimeter & Request Security

- **Web Application Firewall (WAF):** deployed in front of all endpoints to block DoS/abnormal traffic patterns and detect injection attempts (SQLi, XSS, command injection) before they reach the backend
- **Anti-DoS filtering:** blocks randomized or abnormal request patterns targeting the platform
- **Brute-force protection:** rate-limits and locks out repeated failed attempts on login, OTP, and other sensitive endpoints

Input Handling

- **Input validation & sanitization:** enforced on every endpoint, on both frontend and backend, using strict allow-listed formats per field
- **Safe rendering:** user-supplied data is sanitized/escaped before storage or display, and malformed payloads are rejected early

SSRF & Open Redirect Restrictions

- **SSRF protection:** outbound server-side requests are restricted to an allow-list of trusted destinations, with internal/private IP ranges blocked
- **Open redirect protection:** redirect targets are restricted to an allow-list of known-safe URLs; user input never determines a redirect destination directly

Authentication & Session Security

- **Multi-Factor Authentication (MFA):** required at login and enforced again as a transaction-level confirmation, using the password set at registration
- **Password hashing:** credentials are never stored in plaintext
- **Session/token expiry:** authentication tokens expire after a short window (e.g., 5 minutes), forcing re-authentication
- **Transaction rate limiting:** capped at 5 transactions per hour per user to limit the impact of fraud or abuse

3. About the Team

The project is developed by a multidisciplinary team of eight engineers spanning cybersecurity, machine learning, full stack development, DevOps, and penetration testing.

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